Yash Mehta

AI/ML Engineer | Deep Learning Developer | Applied AI Researcher

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience developing machine learning models, deep learning architectures, predictive analytics solutions, and AI-powered applications. Proficient in Python, TensorFlow, Scikit-learn, NLP, computer vision, and time-series forecasting. Developed 10+ technical projects involving LSTM, GRU, CNN, Vision Transformers, and traditional machine learning algorithms. Co-authored research publications in Springer Nature and IEEE proceedings, with research presented at IIT Roorkee's SocProS-25. Interested in developing intelligent, scalable AI solutions for real-world applications.

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript

Machine Learning: Scikit-learn, Random Forest, Feature Engineering, Model Evaluation, Classification, Predictive Modeling

Deep Learning: TensorFlow, CNN, LSTM, GRU, Vision Transformers (ViT)

NLP: TF-IDF, Sentiment Analysis, Text Classification

Time-Series Forecasting: ARIMA, Prophet, LSTM, GRU, Transformers

Computer Vision: OpenCV, Image Processing, Visual Classification

Data Science: Pandas, NumPy, Data Analysis, Data Visualization

Frameworks & Tools: Flask, Jupyter Notebook, Google Colab, Gemini API, Git

Databases: MySQL, MongoDB

SELECTED AI/ML PROJECTS

Amazon Stock Price Prediction | Time-Series Forecasting | 2025

- Developed forecasting models using LSTM, ARIMA, GRU, and Transformer architectures for financial time-series prediction.
- Achieved an RMSE of 0.03285 through model experimentation and evaluation.
- Designed a custom TimeSeriesTransformer supporting multivariate time-series inputs.

Fake News Detection Using Machine Learning | 2025

- Developed a Flask-based fake news detection application integrating TF-IDF, Random Forest, and sentiment analysis.
- Achieved 87% validation accuracy.

- Presented the research at SocProS-25, IIT Roorkee.
- Research subsequently published in Springer's Real-World Applications of Machine Learning.

Fog Detection Using Hybrid Deep Learning Models | 2024-2025

- Developed a hybrid deep learning approach combining CNN and Vision Transformer architectures using the FRIDA2 dataset.
- Reported a 15% improvement in accuracy on real-world haze datasets.
- Integrated OpenCV for image processing and inference.
- Published research in IEEE ICCCA proceedings.

Depression Detection via Facial Cues | 2024

- Developed a CNN and OpenCV-based facial image classification prototype.
- Achieved 82% classification accuracy in project evaluation.
- Integrated the model into a Flask-based demonstration application.

FIR Legal Assistant - LexHack | 2025

- Developed an AI-assisted FIR lodging application using Flask, Google Maps API, and Gemini AI.
- Integrated location-aware functionality and AI-driven assistance.
- Received the Innovation & Impact award from the hackathon evaluation panel.

INTERNSHIPS & RESEARCH

Research Project - Fake News Detection | IIT Roorkee | 2025

- Developed an NLP and machine learning pipeline using TF-IDF, Random Forest, and sentiment analysis.
- Implemented an interactive Flask application.
- Presented research at SocProS-25; subsequently published through Springer Nature.

Software Intern | MNNIT Allahabad | 2024

- Developed REST APIs for academic services using Flask.
- Improved backend throughput by 20%.
- Collaborated through Agile development workflows.

Research Project - Blockchain in Autonomous Vehicles | Bennett University | 2024

- Simulated blockchain protocols for autonomous vehicle communication.
- Reported a 35% reduction in spoofing risks in experimental environments.

RESEARCH PUBLICATIONS

1) Fake News Detection Using Machine Learning and Sentiment Analysis Integrated in a Flask Web Application. In: Pant, M., Deep, K., Nagar, A.K. (eds.), Real-World

Applications of Machine Learning. Springer, Singapore, 2026. DOI: https://doi.org/10.1007/978-981-95-7134-5_23

2) Fog Detection Using Hybrid Deep Learning Models: A Multi-Modal CNN-LSTM Approach. Mehta et al., IEEE ICCCA, 2025. DOI:10.1109/ICCCA66364.2025.11325690

CERTIFICATIONS

- Fundamentals of Deep Learning - NVIDIA
- IBM Machine Learning Professional Certificate - IBM
- Introduction to Generative AI - Google Cloud
- Generative AI Fundamentals - IBM
- AI Foundations for Business - IBM
- Google Data Analytics Professional Certificate - Google
- Quantum Computing & QKD - IIT Delhi Workshop

EDUCATION

B.Tech. - Computer Science & Engineering

Bennett University | 2022–2026 | CGPA: 8.26

Maharshi Patanjali Vidya Mandir | Class XII - CBSE | 2021 | 92.4%

Maharshi Patanjali Vidya Mandir | Class X - CBSE | 2019 | 85.8%